Handling missing data in ETL

Q1. What are the most common reasons for missing data in ETL pipelines?

- Missing data in ETL (Extract, Transform, Load) pipelines occurs when some records or values are not extracted from the source, are lost during transformation, or fail to load into the target system.
- The most common reasons are:
- 1. Source Data Issues: The source database or file contains empty fields, NULL values, or incomplete records.
- 2. Extraction Errors: Network failures, API errors, database connection problems, or incorrect extraction queries may prevent data from being retrieved.
- 3. Transformation Errors: Incorrect transformation logic, filtering conditions, or data type conversions may cause some records to be removed or values to become NULL.
- 4. Data Quality Problems: Invalid, inconsistent, or incorrectly formatted data may be rejected during processing.
- 5. Loading Failures: Database constraints, insufficient storage, permission issues, or target database errors may prevent records from being loaded.
- 6. Schema Changes: Changes in source column names, data types, or table structures can cause the ETL pipeline to miss or incorrectly process data.
- 7. Scheduling and Timing Issues: If data extraction runs before the source data is updated, newly available records may be missed.
- Example: Suppose an ETL pipeline extracts 1,000 customer records from a database, but only 950 records reach the target database. The missing 50 records could be caused by extraction failures, transformation filters, or loading errors.

Q2. Why is blindly deleting rows with missing values considered a bad practice in ETL?

- Blindly deleting rows with missing values in an ETL (Extract, Transform, Load) pipeline is considered a bad practice because it can lead to data loss, biased analysis, and inaccurate business decisions.

The main reasons are:

1. Loss of Valuable Data: A row may contain important information even if only one field is missing.

2. Biased Results: Deleting rows may remove specific types of records and make the dataset unrepresentative of the actual population.

3. Reduced Dataset Size: Removing too many rows can reduce the amount of data available for analysis.

4. Loss of Business Information: Missing values may represent meaningful information, such as customers who have not provided their phone numbers.

5. Difficulty in Auditing: Deleting records without documenting the reason makes it difficult to identify and investigate data quality problems.

6. Incorrect Business Decisions: Incomplete datasets can produce misleading reports, forecasts, and analytical results.

Example:

Suppose a customer dataset contains 1,000 records, and 100 records have missing phone numbers. If we delete all 100 rows, we also lose customers' names, email addresses, and purchase histories, even though those details may be complete.

Better approach:

Instead of blindly deleting rows, an ETL developer should:

- Identify why the data is missing.
- Fill missing values using suitable methods when appropriate.
- Retain records if the missing field is not essential.
- Delete rows only when justified by business rules.
- Log missing values and monitor data quality.

Q3. Explain the difference between:

- Listwise deletion
- Column deletion
- Also mention one scenario where each is appropriate

- ## 1. Listwise Deletion
- Listwise deletion is a method of handling missing data in which the entire row is deleted if one or more required values are missing.
- Example:

Employee ID	Name	Salary
101	Amit	25,000
102	Priya	Missing

103	Rahul	30,000

- After listwise deletion:

Employee ID	Name	Salary
101	Amit	25,000
103	Rahul	30,000

- Appropriate scenario: When analyzing employee salaries, if salary is essential and incomplete records cannot be used for the analysis, deleting those rows may be appropriate.

2. Column Deletion

- Column deletion is a method in which an entire column is removed when it contains too many missing values or is not useful for the analysis.
- Example:

Employee ID	Name	Salary	Middle Name
101	Amit	25,000	Missing
102	Priya	28,000	Missing
103	Rahul	30,000	Missing

- If the middle-name column is unnecessary and mostly empty, it can be removed.

Employee ID	Name	Salary

101	Amit	25,000
102	Priya	28,000
103	Rahul	30,000

- Appropriate scenario: When a column, such as an optional middle name, has mostly missing values and is not required for business analysis.
- **Key Differences**

Basis	Listwise Deletion	Column Deletion
What is removed?	Entire rows	Entire columns
Main reason	Required values are missing	A column has excessive missing values or little value
Main risk	Loss of useful records	Loss of useful information
Example	Removing a record with a missing required salary	Removing an unnecessary column with mostly missing values

- Conclusion: Listwise deletion removes incomplete records, whereas column deletion removes entire fields. In ETL pipelines, both methods should be applied only after evaluating business requirements and the impact on data quality.

Q4. Why is median imputation preferred over mean imputation for skewed data such as income?

- Median imputation is preferred over mean imputation for skewed data because the median is less affected by extreme values (outliers), whereas the mean can be significantly influenced by very high or very low values.

Example

Suppose the monthly incomes of five people are:

₹20,000, ₹25,000, ₹30,000, ₹35,000, ₹2,00,000

- Mean income: ₹62,000
- Median income: ₹30,000

The very high income of ₹2,00,000 increases the mean, making it less representative of the typical income.

If another person's income is missing:

- Mean imputation: Replace the missing income with ₹62,000.
- Median imputation: Replace the missing income with ₹30,000.

The median provides a more representative replacement for this right-skewed income data.

Why is median imputation preferred?

1. Resistant to outliers: Extreme incomes do not significantly affect the median.
2. Suitable for skewed data: Income distributions are often right-skewed.
3. More representative: The median often reflects the typical value better than the mean.
4. Reduces distortion: It avoids using an unusually high or low mean to fill missing values.

Q5. What is forward fill and in what type of dataset is it most useful?

- Forward fill is a method of handling missing data in which a missing value is replaced with the most recent previous valid value in the same column.

Example

Suppose we have the following dataset of daily temperatures:

Date	Temperature (°C)

1 Oct 30

2 Oct Missing

3 Oct Missing

4 Oct 32

5 Oct Missing

After applying forward fill:

Date	Temperature (°C)
1 Oct	30
2 Oct	30
3 Oct	30
4 Oct	32
5 Oct	32

Each missing value is replaced by the last available temperature.

Which datasets are most suitable for forward fill?

Forward fill is most useful for time-series and sequential datasets, where previous values may reasonably remain valid for a short period.

Examples include:

1. Stock market data: Carrying forward the latest available price when appropriate.
2. Sensor data: Filling brief gaps in temperature or humidity readings.
3. Financial records: Carrying forward account balances until the next update.
4. Customer or transaction data: Filling down repeated customer IDs or category labels in structured records.

Advantages

- Simple and easy to implement.
- Preserves the order of observations.
- Useful when values remain stable between updates.

Q6. Why should flagging missing values be done before imputation in an ETL workflow?

- Flagging missing values before imputation means identifying and marking missing or NULL values before replacing them with estimated or existing values.
- It is important because imputation hides the original missing values by filling them in, making it difficult to distinguish actual data from replaced data later.
- ## Reasons for flagging missing values before imputation
- 1. Preserves Data Quality Information: It records which values were originally missing.
- 2. Improves Data Transparency: Analysts can distinguish original values from imputed values.
- 3. Supports Accurate Analysis: It helps analysts understand whether missing data affects the results.
- 4. Enables Better Auditing: Data teams can track how many values were missing and how they were handled.
- 5. Helps Evaluate Imputation Methods: Analysts can check whether the chosen method is appropriate and monitor its impact.
- ## Example
- Suppose an employee dataset contains missing salary values.

Employee	Salary	Missing Flag

Amit	25,000	No
Priya	Missing	Yes
Rahul	30,000	No

- After imputing Priya's missing salary with the median salary:

Employee	Salary	Missing Flag
Amit	25,000	No
Priya	27,500	Yes
Rahul	30,000	No

- The flag remains `Yes` even after the salary is filled, showing that ₹27,500 was imputed rather than originally recorded.

Q7. Consider a scenario where income is missing for many customers. How can this missingness itself provide business insights?

- Missing income data can provide valuable business insights because the absence of information may reveal customer behavior, privacy concerns, or problems in the data collection process.
- **How can missing income provide business insights?**
- 1. Customer Privacy Concerns: Customers may avoid sharing income information because it is sensitive or personal.
- 2. Customer Segmentation: Missing income may be more common among certain customer groups, helping businesses identify differences in customer behavior.

- 3. Data Collection Issues: A high number of missing income values may indicate problems with registration forms, surveys, or data integration.
- 4. Improved Marketing Strategies: Businesses can design better customer surveys and personalize marketing strategies without making unsupported assumptions about customers' income.
- 5. Risk Assessment: Banks and financial institutions can identify gaps in customer information that may need to be addressed before evaluating loan eligibility.
- **Example**
- Suppose an online shopping company has 10,000 customers, and 4,000 customers have not provided their income details.
- The company discovers that most missing values come from customers who registered through mobile applications.
- This may indicate that the mobile registration form does not clearly request income information or that customers prefer not to share it through the app.
- Business action: The company can review its registration process, improve the form, and make income disclosure optional where appropriate.